

Agentic Graph Reasoning

Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

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Thesis Defense

The problem

Language models answer just as fluently when they **do not hold the fact**.

A *hallucination* is output with **no source the model can point to**.

Factually wrong

It contradicts the world (“*Ada Lovelace was born in Paris*”). To catch it you need the true answer already, which is the thing we are trying to produce.

Unsupported

It may even be true, but nothing the system retrieved says so. To catch it you need **only what was retrieved**, and we have that.

A retrieval system can only police the second kind. That is the one this thesis attacks, and the only one it claims.

Where it comes from, and what we can change

Not a bug that escaped testing. It comes from how these models are built, so the response has to be architectural.

The training data

Errors and thin coverage, reproduced faithfully. Fixing them means retraining.

The model

Knowledge is spread across the weights with **no index**. Nothing separates a memorised fact from a plausible continuation.

The prompt

What enters the context window. It is the one origin a deployer controls.

Control what goes in, then check what comes out against it, before anything is emitted.

Knowledge graphs, and why multi-hop is hard

Facts are stored as **triples** of head, relation and tail.

(Ada Lovelace, place_of_birth, London)

Every fact has an address, so a claim can be checked by looking for an edge.

A question that chains two facts is **multi-hop**. Each hop starts from the last, so a wrong first hop is never revisited.

The complication: mediators

Freebase stores a fact with more than two participants as an unnamed **node**. “Rainn Wilson played Dwight in The Office” is one. 63.7% of this graph’s nodes are mediators, so a path that reads as two hops is often three.

Nothing in the graph is called plays. The relation a question names is rarely the relation the graph stores.

Where existing approaches stop

Approach	What it does	Where it stops
Parametric	Answers from weights	No external evidence at all
Vector-RAG	Retrieves text once, then reasons	Retrieval happens <i>before</i> reasoning begins
Static GraphRAG	Retrieves a fixed neighbourhood	Radius fixed before the question is understood
Agentic navigation	Interleaves retrieval and reasoning	Emits whatever the last generation call produces

Agentic systems already name only what they traversed. Their problem is **precision**. A real, traversed entity can still be the wrong answer.

None of the four checks what the answer asserts *before* it is delivered.

What guided beam search is

It keeps only the most promising branches:

1. List every edge out of the current **frontier**.
2. **Score** each for how likely it is to help.
3. Keep the best b , the **beam width**.
4. Expand those, and repeat to a depth cap.

Guided means the score is a judgement.

Why bound it

Entities carry hundreds of relations, so three unbounded hops is millions of paths. When the scorer is a **model call**, the width *is* the cost, and it multiplies with every hop.

Everything outside the beam is discarded unseen, and a plain beam has no way back for it.

What everyone else does

System	Exploration	Decomp.	Backtrack	Verifies first
GraphRAG	Fixed radius, no search	No	No	No
RoG	Generated relation paths	Partial	No	No
Think-on-Graph	LLM-guided beam search	No	No	No
Plan-on-Graph	Adaptive planning	Yes	Yes	Partial

RoG's answers are grounded by construction, but nothing examines them. Chain-of-Verification and FActScore check claims *after* generation. Published Hits@1 is higher and *not* comparable, because of other backbones, subsets and full Freebase.

No surveyed graph-navigating system checks its answer's claims against the triples it retrieved, before answering.

Research questions

RQ1 Does agentic navigation improve multi-hop factual accuracy, and does the advantage **grow with hop count**?

RQ2 What does pre-generation verification **contribute** beyond graph navigation?

RQ3 Which components contribute what, at what **token cost**?

RQ2 asks what verification *contributes*, not whether it reduces hallucination. The answer has several parts.

Six contributions

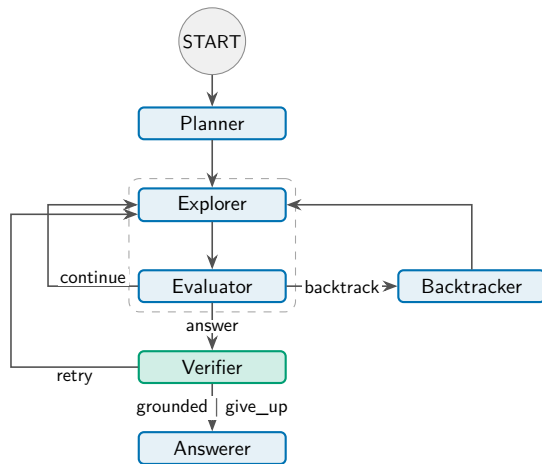
1. AGR and its verification layer
2. A component-level ablation
3. Stratum-dependent decomposition
4. The echo attractor
5. Benchmark-defect rates
6. A pre-specified protocol

AGR: An explicit state machine

Six nodes. The model is consulted only at decision points.

- **Planner**: sub-objectives
- **Explorer**: best-first, beam 3
- **Evaluator**: objective met?
- **Backtracker**: best earlier frontier, ban list
- **Verifier**: checks claims first
- **Answerer**: emits what survived

Three cycles (*continue*, *backtrack*, *retry*), all bounded by budgets rather than by model behaviour.



Constrained tools, not free-form queries

One agent with six nodes, and only three of them reach the graph, each through a fixed operation.

Operation	Node	Returns	At most
search_entity	Planner	Surface form $\rightarrow$ node	3 candidates
get_relations	Explorer	Candidate relations	≤ 300 to the scorer
get_neighbors	Explorer	Neighbours of an entity	≤ 200 per expansion
verify_connection	Verifier	Adjacency, full graph	one boolean

Evaluator, Backtracker and Answerer read state only. The agent never writes Cypher, and every call is logged with its arguments and result.

Determinism in the tools is what makes the verification layer possible.

The Structural Verification Layer

Before anything is emitted:

1. The draft is split into **atomic claims**
2. Each claim is checked against the graph, the **triples actually traversed** first
3. Claims grounded by the traversal keep those triples as **evidence**
4. Unsupported claims send the agent back to **re-explore**, or are **dropped**

The behaviour this buys

Where it cannot ground a claim, the system **hedges rather than asserts**. Where the walk grounds it, the answer arrives with that evidence.

The layer this thesis set out to contribute. Measured, it buys auditability rather than accuracy.

What *structural* means, and what it does not

The graph decides. The model is consulted only where the graph is silent, and it never overrides the graph.

What it does not mean

It tests *adjacency*, not the relation the claim names, and not its direction. A *mother* claim passes on a *child* edge.

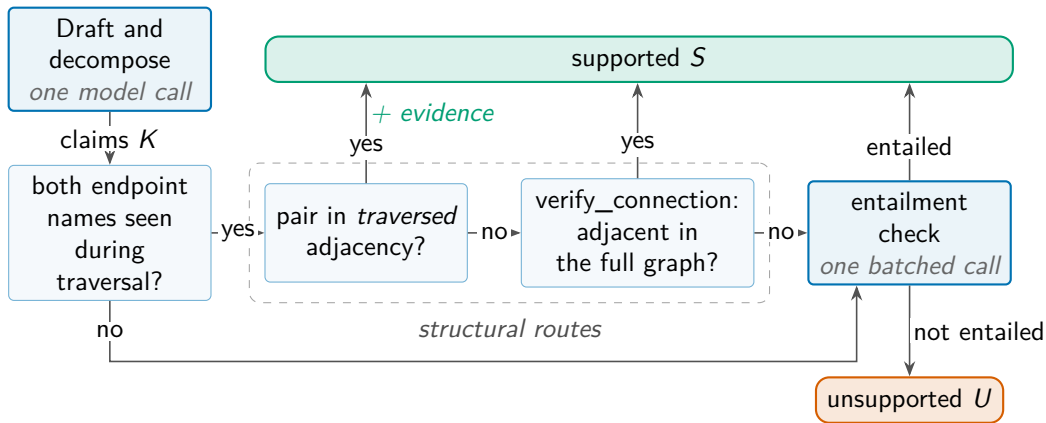
What it cannot catch

A claim can be *true* and still be the wrong answer. That is the echo attractor of slide 27.

What “its evidence” means

Only the check against the *walked* graph attaches triples. The other two routes accept with nothing attached. The log keeps the *count*, not the triples.

One claim, three routes



Only the third route costs a model call. Neither lookup reads the relation. The node exits grounded if U is empty, else retry while budget remains, else give_up.

One question, end to end

WebQTest-960, “who plays dwight in the office”

1. **Planner**: two sub-objectives. The given mention “The Office” resolves to one node.
2. **Explorer**: keeps three relations. The best is `tv.tv_program.regular_cast`
3. **Evaluator**: objective met, resolves Dwight Schrute
4. **Explorer**: keeps three again. The best is `tv.regular_tv_appearance.character`
5. **Evaluator**: resolves Rainn Wilson, decides answer
6. **Verifier**: 2 claims, 2 supported, none dropped

What came out

“Rainn Wilson plays Dwight Schrute in The Office.”

6 model calls · depth 2 · at most 18 supporting triples

Both hops run *through* a mediator. The relation named is the edge *into* the appearance record, not the one that lands on the entity.

The environment and the question sets

Freebase-derived graph

Property	Value
Entities	2,592,892
Triples	8,309,194
Distinct relations	7,058
Import time	36.4 s

A union of RoG's per-question subgraphs, each cut to contain its answer, so friendlier than full Freebase.

Certified question sets

	WebQSP	CWQ
Questions	400	400
Gold (median)	1.5	1.0
Reachable	97.0%	99.2%
Multi-hop	132	260

Reachability caps every Hits@1, for entity answers only. Dates, quantities and superlatives are beyond this graph.

WebQSP and ComplexWebQuestions (CWQ), unmodified. Absolute levels are optimistic, and every system faces the same graph.

Making the comparison fair

Five systems, **one frozen backbone**,
one budget:

- No-retrieval (parametric control)
- Vector-RAG
- GraphRAG (static, one hop)
- Think-on-Graph (reimplemented)
- **AGR**

The graph systems start from the datasets' topic entities. The control is there because these benchmarks may be partly **memorised**.

Held constant

gpt-5.4-mini-2026-03-17 at temperature 0,
the 25-call cap, the questions and the graph.

Not held equal

ToG prunes to 40/20 candidates per step,
AGR to 300/200. **Narrower is cheaper**, so it
cannot explain ToG's clipping, but its
unclipped score is a **lower bound**. The
baselines' prompt is plain, while AGR's carries
grounding rules.

Differences are attributable to architecture, not to model capacity. Both exceptions are on this slide.

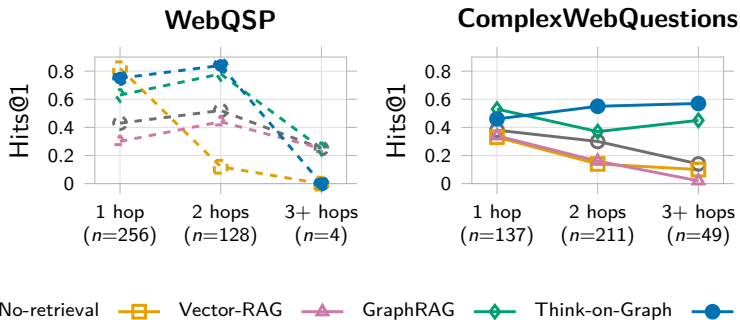
Main results

System	WebQSP		CWQ		Cost / q	
	Hits@1	F1	Hits@1	F1	Tokens	Calls
No-retrieval	0.453	0.305	0.307	0.279	113	1.0
Vector-RAG	0.568	0.432	0.203	0.168	527	1.0
GraphRAG	0.340	0.262	0.205	0.183	531	1.0
Think-on-Graph	0.660	0.477	0.435	0.378	3,615	12.8
AGR	0.755	0.642	0.522	0.469	4,511	6.2

Cost is WebQSP. On CWQ, AGR spends 6,818 tokens and 8.9 calls, Think-on-Graph 5,572 and 18.2. All eight paired McNemar tests favour AGR ($p < 0.004$), and the 95% Hits@1 intervals are ± 4 to 5 points.

Ahead of every baseline, at half ToG's calls and 22 to 25% more tokens.

RQ1: Does agentic navigation improve multi-hop factual accuracy?



On CWQ, AGR goes $0.46 \rightarrow 0.55 \rightarrow 0.57$, the **only** system ending above where it started. Three of the other four decay monotonically. WebQSP is dashed ($n=4$ at 3+).

Yes. The evidence is a shape, not a difference of means.

The caveat I want to raise myself

Hits@1 on...	WebQSP		CWQ	
	ToG	AGR	ToG	AGR
questions ToG <i>finishes</i>	0.852	0.788	0.629	0.607
questions ToG is <i>clipped</i> on	0.197	0.675	0.188	0.415
clip rate	29.2% (117)		44.0% (176)	

Where Think-on-Graph finishes inside the shared 25-call cap, it is **ahead of AGR on both datasets**. The entire aggregate margin comes from the questions it cannot finish, 117 and 176 of 400. Beam search pays width times depth in calls, and AGR's plan and scored frontier do not multiply.

The claim is efficiency under a fixed budget, not better reasoning per step.

RQ2: What does pre-generation verification contribute beyond graph navigation?

System	Asserted	Ungrounded
No-retrieval	1,001	221 (22.1%)
Vector-RAG	1,329	45 (3.4%)
GraphRAG	1,024	8 (0.8%)
Think-on-Graph	1,265	0 (0.0%)
AGR	1,709	0 (0.0%)

Both datasets pooled. A stricter semantic check, on a sample of each system's answers, supports only 36.7 to 66.7% of them.

Think-on-Graph reaches 0.0% too. So this is a property of *graph navigation*, not of the verification layer.

RQ2: What verification does *not* do

It does not move accuracy

Remove it and Hits@1 and F1 do not detectably change, at $p = 1.0$ on *both* datasets. It passed 77 of 80 development drafts untouched and changed 2 of 398 test answers, one each way.

Its evidence ends with the run

The log keeps the **count** of supporting triples and drops the list itself.

It does not explain precision

On CWQ, AGR is right on 67.9% of what it asserts, Think-on-Graph on 56.1%. Removing the layer moves AGR's precision by a point at most. The lead belongs to the architecture, and possibly in part to its drafting prompt.

Not a retraction of slide 17. That was AGR against four baselines. This is AGR against *itself* with one part removed. **What it does buy is the next slide.**

RQ2: So what does it do?

It withholds the ungrounded

Remove it and hedging falls

23.2% $\rightarrow$ 20.2% on CWQ. Those six answers were all wrong. It falls 8.5% $\rightarrow$ 8.0% on WebQSP, one answer, which was right.

It attaches evidence

From **one route of three**, it attaches supporting triples, inside the run only.

The other two routes, `verify_connection` and entailment, attach none.

It rejected a fabricated actor

A draft said Marlon Brando was in *Joy*.

He was not, and the verifier rejected the claim. De Niro was, and the evaluator had resolved him.

Not right more often. Right **for a reason it can show**, at a small cost in recall.

Does it just refuse more often?

System	WebQSP	CWQ
No-retrieval	12.2%	38.0%
Vector-RAG	27.0%	48.2%
GraphRAG	55.8%	60.8%
Think-on-Graph	18.8%	22.5%
AGR	8.2%	23.0%

A hedge scores zero on Hits@1, exactly as a wrong answer does. AGR hedges **least of the five** on WebQSP, where it is also the most accurate. These are the full 400 questions per dataset. The previous slide's 8.5 and 23.2 are the ablation half-samples.

Not more often. The lowest rate of the five on WebQSP, and on CWQ within half a point of Think-on-Graph while more accurate.

RQ3: Which components contribute what, at what token cost?

Condition	WebQSP		CWQ		Tokens
	F1	p	F1	p	
Full system	0.659	—	0.445	—	4,562
No planner	0.742	0.006	0.398	0.088	3,159
No backtracking	0.646	0.727	0.445	1.000	4,261
No verifier	0.663	1.000	0.436	1.000	4,460
Embedding-only scoring	0.643	0.664	0.437	0.481	3,413

p is a paired McNemar test on Hits@1, $n = 200/198$, uncorrected. *Full system* is the main run on the same halves, and *embedding-only* drops the model term. Tokens are for WebQSP.

Three of the four change nothing detectable. The fourth is on the next slide, and its sign is backwards.

RQ3: One effect, and its sign is backwards

Removing the planner *improves* WebQSP

+0.083 F1, $p = 0.006$, and -31% tokens. CWQ trends the other way, -0.047 at $p = 0.088$. Without the planner, WebQSP rises at one hop ($0.75 \rightarrow 0.85$) and at two ($0.84 \rightarrow 0.89$), while CWQ's two-hop stratum falls $0.48 \rightarrow 0.34$.

How decomposition hurts

It can strip context. “This 33rd president was the nation’s leader during WW2” became two lookups, WW2 was never used, and the run answered Woodrow Wilson. Undecomposed, it found Truman in three calls.

Gate decomposition on structure. The measurement forced that conclusion rather than confirming one.

Every failure, read and labelled

All 256, read and labelled by hand.

Category	n
Relation selection	64
Composite claim	47
Knowledge-graph gap	43
Decomposition error	38
Answer selection	15
Echo attractor	13
Six smaller categories	36

Totals hide the shape

The 256 are AGR's failures over the 800 test questions, less the benchmark defects adjudicated before the census and the surface-form near-misses. Wrong and hedge are *never pooled* in the thesis. The shape flips too. Composite claim is 1 on WebQSP against 46 on CWQ.

The largest category is **relation selection**, the agent taking the wrong edge rather than inventing one.

The characteristic error of a graph navigator is *not* invention.

The echo attractor

Sixth in that census, with 13 cases, and the one worth naming. A real, grounded entity **one hop** from the answer.

Asked who plays Lex Luthor on *Smallville*, AGR named John Glover, who plays his father Lionel.

Different systems fall into it **together**, so no evaluation treating them as independent can see it. A policy of rescoring on majority agreement turns it into apparent **correctness**.

Why it matters

Any grounding check *passes* it, since it is true. The entity is real, it was traversed, and the triple exists. By construction, verification cannot catch it.

The contribution is the mechanism, named, not the count.

The three answers

- RQ1 Yes.** AGR leads every accuracy cell, all eight paired tests reject, and on CWQ only AGR rises with hop count. The margin over Think-on-Graph lies in the budget.
- RQ2** Navigation, not verification, removes structural hallucination. The precision lead belongs to the architecture. The layer buys auditability, not accuracy.
- RQ3** Only the planner has a detectable effect, and its sign depends on the stratum. The whole system runs at half the agentic baseline's calls.

Navigation is where the accuracy comes from. Verification is where the auditability comes from. Decomposition is sometimes harmful.

Contributions

1. AGR and its **Structural Verification Layer**
2. A **component-level ablation** of four mechanisms
3. **Stratum-dependent decomposition**, gated on structure
4. The *echo attractor*, named
5. **Benchmark-defect rates** for 57 questions with wrong or ambiguous gold
6. A **pre-specified** protocol of four decisions, one missed ($\kappa = 0.6995$ against 0.7)

What I would not claim

- The verifier logs only what it **rejects**. Wrongful acceptance is unmeasured, and the evidence is not persisted.
- One environment, one backbone, one run
- Think-on-Graph leads where it finishes, from a **narrower candidate set** (40/20 against 300/200)
- The title's terms at face value

What comes next

Repairs the census earned

- Tell the decomposer which argument is the answer (9 extraction-bug cases)
- Log accepted claims with the triples that matched them
- Re-run the baselines at equal candidate widths and a wider radius

Operators the failures specify

- Gate decomposition on question structure
- A set-intersection operator (47 composite-claim cases)
- Ordinals and literal values (43 knowledge-graph-gap cases)

Fact verification against a knowledge graph is the easier half. Relevance verification is the half still open.

Thank you

Questions

Backup: budget configuration

Parameter	Value	Enforced in
max_depth	4	both routers
beam_width	3	explorer (slice)
max_anchors	5	explorer <i>and</i> evaluator
max_backtracks	3	post-evaluation router
max_llm_calls	25	LLM client
max_verify_iters	2	post-verification router, verifier
max_seconds	300	both routers

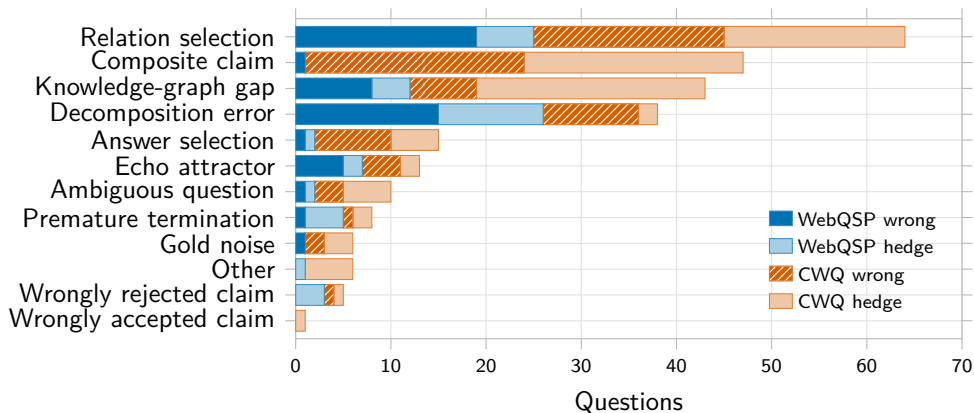
Termination does not depend on model behaviour. Every cycle passes through a router that checks a monotone counter.

Backup: which budgets actually bind

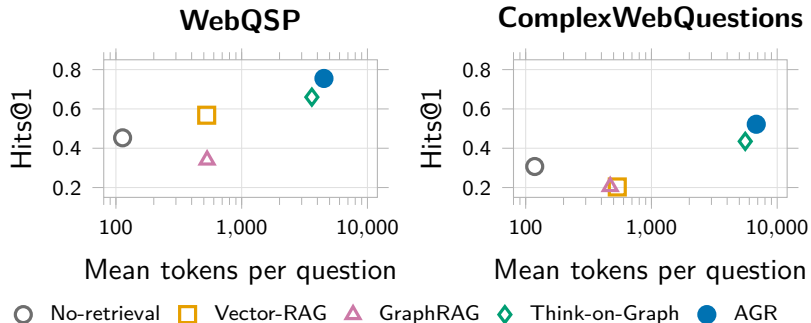
Budget	WebQSP	CWQ	Pooled
Depth cap	16.5%	34.8%	25.6%
Backtrack cap	3.2%	9.2%	6.2%
Verify-iteration cap	1.0%	2.8%	1.9%
Call cap	0.0%	0.0%	0.0%

Percentage of questions on which each budget refused a request. AGR never reaches the 25-call cap. That is what makes a clipped Think-on-Graph a meaningful comparison.

Backup: the full failure census



Backup: accuracy against cost



On *tokens* Think-on-Graph is **not** dominated. It buys lower accuracy at a lower token price. On *calls*, which the budget meters, it is.

The frontier depends on which cost you charge.

Backup: the benchmark was wrong 57 times

The same cross-system agreement is also a **detector**. Consensus flagged 105 questions and adjudication confirmed 41. The gap is the attractor above, not label noise.

- 41 excluded before the census
- 16 more found inside it, as gold-noise or ambiguous rows
- No question counted in both

⇒ 57 **distinct questions**

Why report it

Reported defect rates for these two standard benchmarks are rare and usually anecdotal. This is a counted rate with a documented adjudication procedure and per-question provenance.

A contribution that outlives this particular system.

Backup: AGR against RoG

System	WebQSP		CWQ	
	Hits@1	F1	Hits@1	F1
RoG (published)	85.7	70.8	62.6	56.2
AGR (this work)	75.5	64.2	52.2	46.9

RoG leads on all four, and the gap is outside AGR's confidence intervals. But RoG is fine-tuned on the training splits of both benchmarks, 2,826 and 27,639 questions, while AGR trains on nothing. RoG reports over the full test splits, AGR over the certified 400-question samples of those splits.

Same data, different class of system, supervised against zero-shot. Worth stating, but it is not an architecture comparison, and no protocol here could make it one.

Backup: what changed from the proposal

Proposal	Now	Why
“MCTS-inspired” search	Withdrawn	No rollouts and no value backpropagation. It is guided best-first search with a bounded beam and backtracking.
FactBench as graph and judge	Dropped	An evaluator whose facts sit inside the retrieval graph cannot measure hallucination.
Path fidelity metric	Deferred	Needs gold SPARQL relation chains, which the RoG distribution does not carry.
A registered protocol	Pre-specified	Four decisions were fixed in writing, but nothing was filed with a registry.